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Introduction to p-adic Numbers
Homework Sheet 3

Due: November 4

Recall from the script that a *metric* $d\colon X \times X \to \mathbb{R}$ is a function for a set X satisfying for all $x, y, z \in X$:

(i) $d(x, y) = 0$ iff $x = y$;

(ii) $d(x, y) = d(y, x)$;

(iii) $d(x, y) \leq d(x, z) + d(z, y)$.

A sequence $(x_i)_{i\in\mathbb{N}}$ in X is called *Cauchy* for d if for every $\varepsilon > 0$ there exists $N > 0$ such that $d(x_i, x_j) < \varepsilon$ for all $i, j \geq N$. Two metrics d, d' on X are called *equivalent* (denoted by $d \sim d'$), if any sequence $(x_i)_{i\in\mathbb{N}}$ in X is Cauchy for d iff it is Cauchy for d'. By definition, two norms on X are equivalent iff the respective metrics are equivalent.

Exercise 3.1

Equivalences of norms.

(a) A norm $|-|\colon \mathbb{Q} \to \mathbb{R}$ defines the metric $d(x, y) := |x - y|$ for all $x, y \in \mathbb{Q}$.

(b) Given a norm $|-|$ and a fixed real positive number $0 < \alpha \leq 1$. Prove that $|-|^\alpha$ is a norm which is equivalent to the norm $|-|$.

(c) In the previous item, if $|-|$ is nonarchimedean, then $|-|^\alpha$ is a norm for any fixed real $\alpha > 0$, and in that case it is equivalent to the norm $|-|$.

Exercise 3.2

Continuity of operations in a field with norm.

Let $\Bbbk$ be a field with a norm $|-|$ which induces the metric d. Prove that the following maps are continuous (with respect to the product metric $d\big((x, y), (x', y')\big) := d(x, x') + d(y, y')$ on $\Bbbk \times \Bbbk$):

(a) $\mu\colon \Bbbk \times \Bbbk \to \Bbbk$, $(x, y) \mapsto xy$.

(b) $\eta\colon \Bbbk \setminus \{0\} \to \Bbbk$, $x \mapsto x^{-1}$.

Exercise 3.3

Nonarchimedean norms.

For a norm $|-|$ on $\mathbb{R}$ suppose that $S := \sup\{|n|\,|\, n \in \mathbb{Z}\} \leq C$ holds for some $C \in \mathbb{R}$. (Here $\mathbb{Z}$ is embedded to $\mathbb{R}$ by unit multiples.)

(a) Prove that $S = 1$ and $|-|$ is nonarchimedean.

(b) Deduce that if a norm $|-|$ on $\Bbbk$ is archimedean, then for any $x, y \in \Bbbk$, $x \neq 0$, exists $n \in \mathbb{Z}$ such that $|nx| > |y|$ holds.

Exercise 3.4

Legendre formula.

Write an integer $n = a_0 + a_1 p + \cdots + a_s p^s$ base p for some s. Let $S_p(n) := \sum_i a_i$ be the sum of the respective digits for n. Recall $\nu_p(n)$ is the maximal degree of p dividing $n \in \mathbb{Z}$. Prove that

$$\nu_p(n!) = \frac{n - S_p(n)}{p - 1}.$$